

Part 4: Comparison

1. AM vs FM Comparison

Feature	AM	FM
Bandwidth	Low	High
Noise Resistance	Poor	Excellent
Signal Quality	Moderate	High
Complexity	Simple	Complex

AM (Amplitude Modulation) is simple and requires less bandwidth, but it is highly affected by noise because the information is carried in the amplitude.

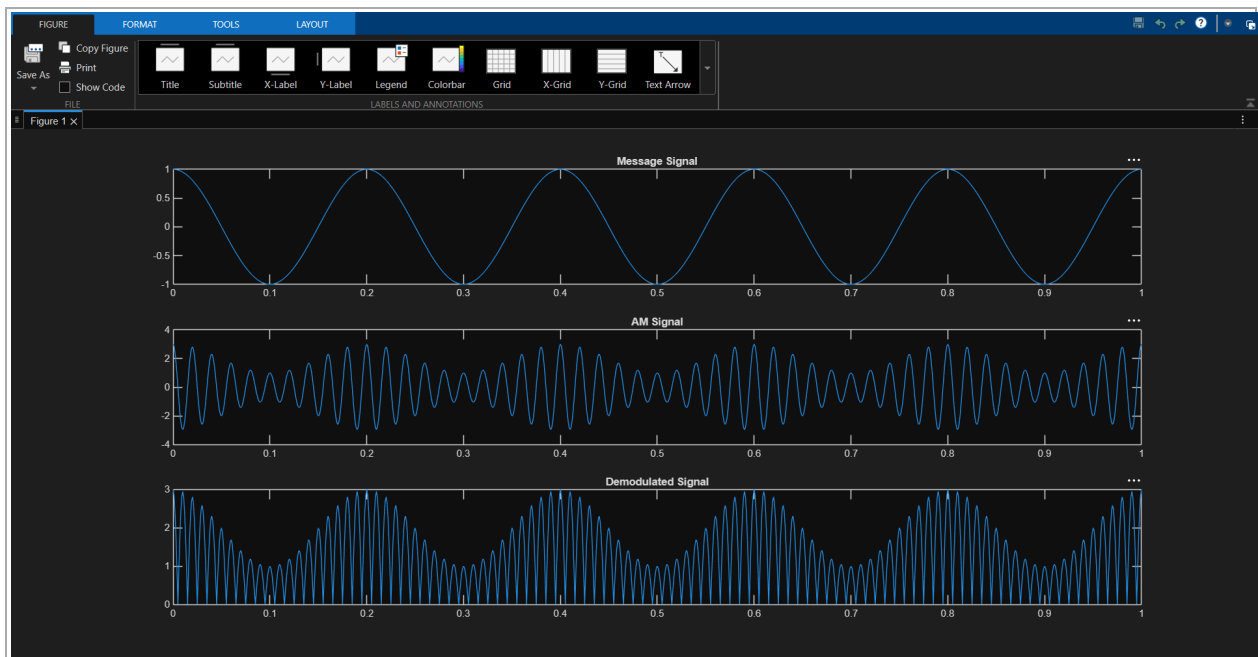
FM (Frequency Modulation) provides better signal quality and is more resistant to noise since the information is carried in the frequency. However, it requires more bandwidth and is more complex to implement.

AM: Easily affected by noise (noise monster)

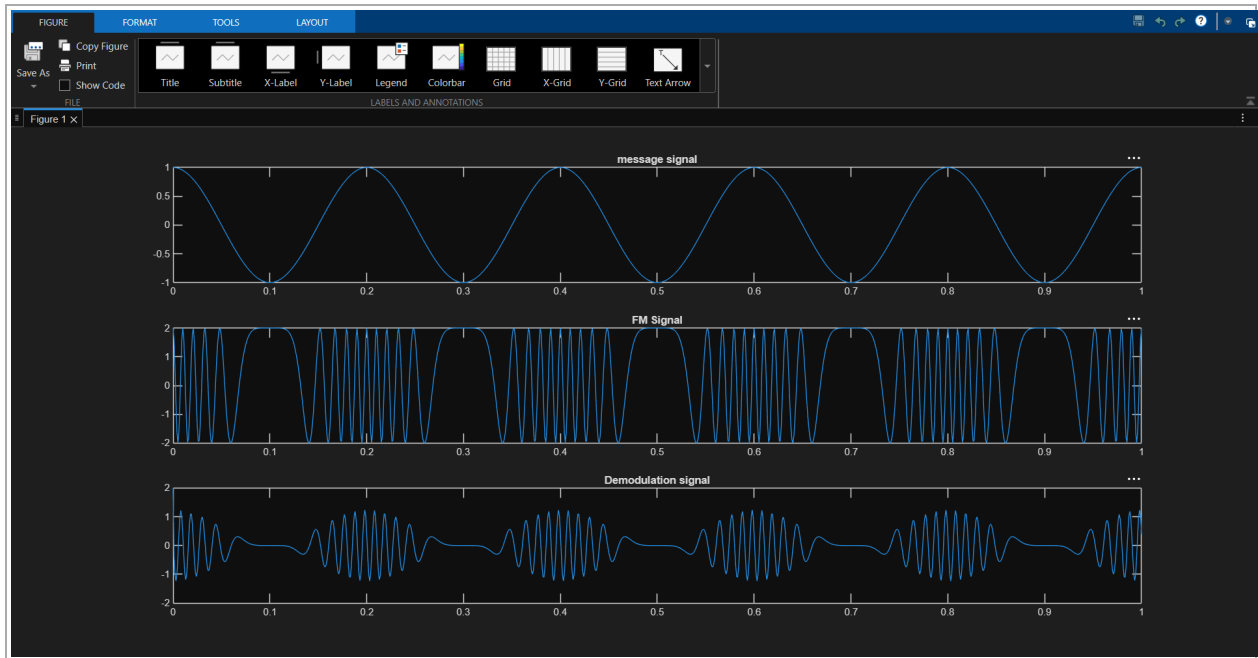
FM: Strong against noise (better)

From the results, FM is preferred in noisy environments, while AM is suitable for simpler and bandwidth-limited systems.

AM Result



FM Result



2. ASK vs FSK Comparison

Feature	ASK	FSK
Bandwidth	Low	High
Noise Immunity	Poor	Good
Bit Error Performance	Worse	Better
Complexity	Simple	Complex

ASK (Amplitude Shift Keying) is simple to implement and requires less bandwidth, but it is highly affected by noise because the information is carried in the amplitude of the signal.

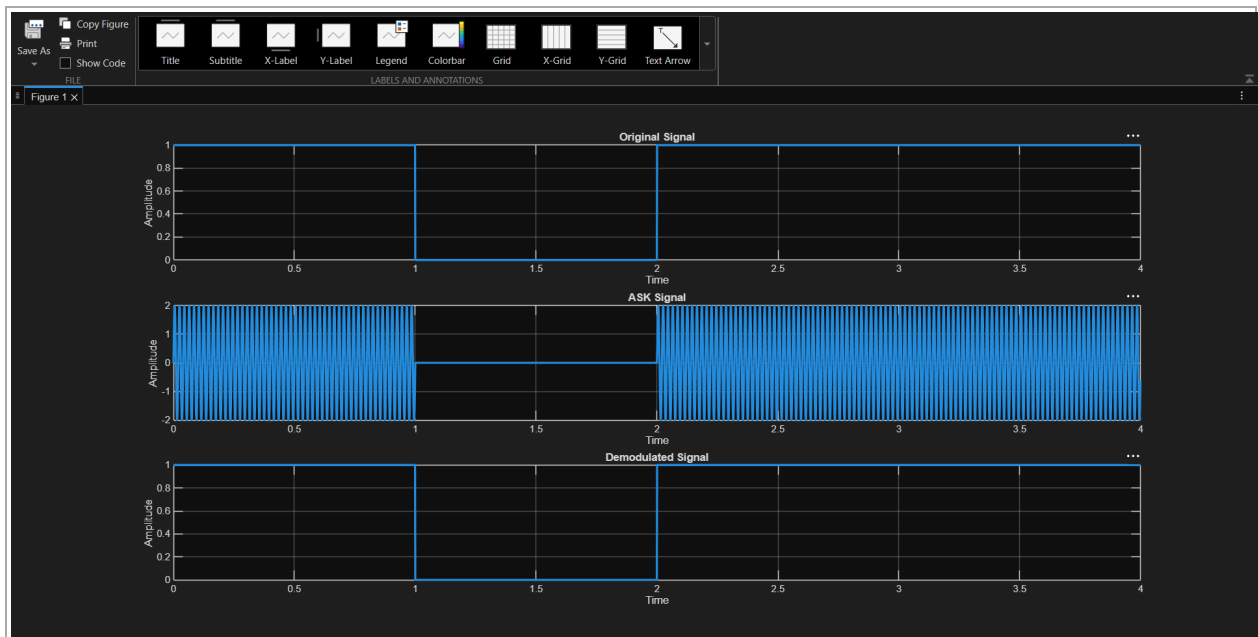
FSK (Frequency Shift Keying) provides better noise immunity since the information is encoded in the frequency rather than amplitude. However, it requires more bandwidth and is more complex to implement.

ASK: Easily affected by noise, simpler to implement.

FSK: More resistant to noise, better bit error performance.

From the results, FSK is preferred in noisy environments due to its better noise immunity and lower bit error rate, while ASK is suitable for simpler and bandwidth-limited systems.

ASK Result



FSK Result

